Local models and difference formulas

Use	Topic	Statement and conditions
Recall	**Local linear model**	$L(a+h) = f(a) + f'(a)h$. Require a finite derivative at a; h is a signed input change. Estimated change: $f'(a)h$.
Understand	**Locality**	$r(h) = f(a+h) - L(a+h)$ satisfies $r(h) / h \to 0$ as $h \to 0$ when f is differentiable at a. No universal finite-step radius or tolerance follows.
Recall	**Error measures**	Estimate A, target v: absolute error $\lvert A - v \rvert$; relative error $\lvert A - v \rvert \;/\; \lvert v \rvert$ only for $v \neq 0$. Percentage error:100 times relative error.
Recall	**Forward difference**	$D_+(h) = \dfrac{f(a+h) - f(a)}{h}$, $h > 0$. Both inputs must be allowed.
Recall	**Central difference**	$D_0(h) = \dfrac{f(a+h) - f(a-h)}{2h}$, $h > 0$. Samples lie on both sides; full separation is $2h$.
Consult	**Backward difference**	$D_-(h) = \dfrac{f(a) - f(a-h)}{h}$, $h > 0$. At a domain endpoint, label a one-sided rate.
Understand	**Existence is a separate question**	Differentiability implies exact difference convergence. Central convergence alone does not imply differentiability: $\lvert x \rvert$ at zero gives $D_0(h) = 0$.

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Checking numerical evidence

Use	Topic	Statement and conditions
Understand	**Exact test case**	For $f(x) = x^3$ at 1: derivative 3; $D_+(h) = 3 + 3h + h^2$; $D_0(h) = 3 + h^2$. These identities are not universal error formulas.
Understand	**Competing effects**	Truncation is finite-step error in exact arithmetic. Rounding/cancellation can dominate at small steps. Equal stored inputs or outputs can destroy the useful difference.
Consult	**Bounded output errors**	If each sampled output error has magnitude at most δ: contribution at most $2\delta / h$ forward/backward, δ / h central. Excludes truncation and other errors.
Consult	**Report a numerical check**	State model/domain, base input/units, formula, steps, independent reference, observed errors, edge cases and limitations. A finite table is not a universal bound.

Consult · Python quick lookup

```
def central(f, a, h):
    if h <= 0:
        raise ValueError("Use a positive step.")
    if a + h == a or a - h == a:
        raise ValueError("A sampled input did not change.")
    return (f(a + h) - f(a - h)) / (2 * h)

def cube(x):
    return x ** 3

for h in [0.1, 0.01, 0.001]:
    estimate = central(cube, 1.0, h)
    print(h, estimate, abs(estimate - 3.0))
# Use finite real inputs; check domain and derivative existence.
# A passed guard does not certify accuracy.
# Restart and run all cells; do not round intermediate outputs.
```

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